

Project Title

Smart Soil Health Card Generator Using IoT

Problem Understanding and Empathy Development

Agriculture remains the backbone of many economies, especially in rural regions where farmers heavily depend on the productivity of their soil for livelihood. However, many farmers lack access to timely and accurate information about their soil's health, leading to inefficient crop production, overuse or misuse of fertilizers, and excessive water usage. Traditional methods of soil testing are time-consuming, expensive, and often inaccessible to small-scale farmers.

Through direct engagement with farmers and agricultural experts, it is evident that the need for a **low-cost, real-time, and automated system** to monitor soil health is critical. Interviews and field visits highlighted common pain points:

- Farmers do not always understand what specific deficiencies their soil has.
- Water is frequently wasted due to unmonitored irrigation.
- Soil pH imbalances are not addressed due to lack of awareness or timely diagnosis.
- Fertility levels of the soil are unknown, affecting crop choice and yield.

By empathizing with farmers and observing their practices, it became clear that an intelligent, localized, and automated solution can significantly improve agricultural outcomes and conserve resources.

Problem Statement:

In many parts of India and globally, farmers lack access to real-time soil health data and actionable guidance. This leads to inefficient irrigation, imbalanced fertilization, and crop loss. Traditional soil health cards are static, delayed, and do not offer automation or real-time decisions.

Farmers require a solution that not only monitors critical soil parameters but also responds to them automatically and provides clear, localized recommendations to improve crop yield and sustainability.

Project Objectives:

1. **Monitor** soil parameters like pH, moisture, and temperature in real time using IoT sensors.
2. **Automate irrigation** when moisture falls below a defined threshold.
3. **Suggest pH correction** methods when soil acidity or alkalinity is detected.
4. **Calculate a fertility index** based on all sensor inputs and recommend crops accordingly.
5. **Provide an easy-to-read Soil Health Card** digitally via a mobile/web interface.
6. **Empower farmers** with insights and alerts to improve decision-making and productivity.

System Components:

Component	Function
ESP32 / NodeMCU	Microcontroller for sensor integration
Soil Moisture Sensor	Monitors water content in soil
Soil pH Sensor	Measures soil acidity/alkalinity
Temperature Sensor (DHT11/22)	Measures ambient temperature
Relay Module	Controls the water pump/solenoid valve
Water Pump / Valve	Used for automated irrigation
IoT Dashboard	Displays live data, recommendations
Cloud Storage	Data logging
Mobile App or Web Portal	Farmer interface for alerts & data

Key Features & Innovations:

Real-Time Monitoring:

- Sensors continuously read **moisture**, **pH**, and **temperature** levels.
- Data is sent to cloud in real-time.

Automated Irrigation:

- When **moisture** < (e.g., 30%), system automatically turns on **pump** using a relay.
- Stops irrigation once moisture is restored to optimal level (e.g., 60%).

pH Level Advisory:

- If **pH** < 6.0: Recommends **lime**, **wood ash**, or **biochar**.
- If **pH** > 7.5: Suggests **elemental sulfur**, **organic compost**, or **acidifying fertilizers**.
- Shown on app in simple language (e.g., "Your soil is alkaline. Add sulfur or organic compost.")

Fertility Index Calculation:

- Uses formula combining:
 - pH balance
 - NPK (if extended)
 - Moisture adequacy

- Soil temperature
- Outputs a score (0–100) and categorizes:
 - 0–40: Poor fertility → Crop suggestion + recovery plan
 - 41–70: Moderate fertility → Minor amendments
 - 71–100: Good fertility → Best crops & care tips

Alerts & Notifications:

- Farmers receive SMS/app alerts:
 - “Irrigation started due to low moisture”
 - “Soil pH is acidic, add lime”
 - “Fertility index: 78 – Suitable for wheat and soybean”

User Interface (UI) Features:

- Real-time dashboard with graphs and alerts
- Visual soil health card (printable)
- Recommendations section
- Historical data trends
- Crop calendar & climate tips

Design Space Understanding:

Layer	Tools/Platforms
Microcontroller	ESP32, NodeMCU
Sensors	Capacitive Moisture Sensor, pH Probe
Cloud	ThingSpeak / Firebase / AWS IoT Core
UI	HTML, CSS, JS + React / Flutter App
Communication	WiFi (via ESP32)
Power	Solar panel + battery (optional)

Benefits:

- **Reduces water usage** through automated irrigation
- **Improves yield** by balancing soil conditions
- **Saves farmer effort and cost** via automation
- **Informs and educates** farmers with easy-to-understand advice
- Encourages **sustainable farming** practices

Conclusion:

Your Smart Soil Health Card Generator is a complete **IoT-driven precision agriculture solution**. By combining real-time monitoring, automation, and advisory intelligence, it empowers farmers to make data-backed decisions for improved productivity and sustainability. It stands as an innovative, scalable, and impactful solution for modern agriculture.

